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Exploring Adolescent and Young Adult Cancer Survivors' Experience with Cancer Treatment-Related Symptoms: A Qualitative Analysis of Semi-Structured Interviews

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Purpose: Few studies have specifically targeted symptom management interventions for adolescent and young adult (AYA) cancer survivors. A greater understanding of AYA cancer survivors' experiences with cancer treatment-related symptoms would help develop age-appropriate oncology symptom management interventions. The purpose of this qualitative analysis was to explore AYA cancer survivors' experience with cancer treatment-related symptoms.
Methods: Nineteen post-treatment AYA cancer survivors (18–39 years old) who self-reported moderate–severe cancer treatment-related symptom severity participated in video conferencing or telephone interviews. The questions in the interview guide queried participants to share their experience with cancer treatment-related symptoms. Inductive content analysis was used to identify themes from the interviews.
Results: The themes that emerged from the interviews included (1) cancer treatment-related symptoms negatively affected AYA cancer survivors' quality of life (e.g., symptoms served as a reminder of cancer recurrence possibility); (2) AYA cancer survivors' attitudes and feelings about communicating cancer treatment-related symptom concerns to clinicians (e.g., patient–clinician communication was bolstered when AYAs perceived that symptoms were being taken seriously); (3) AYA cancer survivors are interested in oncology symptom management clinical trials, but logistical challenges are barriers to participation; and (4) AYA cancer survivors are interested in nonpharmacological treatments for symptom management.
Conclusion: Results highlight the burden of cancer treatment-related symptoms on day-to-day life among post-treatment AYA cancer survivors. Future work is needed to identify nonpharmacological symptom management interventions, strategies to improve patient–clinician communication about symptoms, and strategies to increase the visibility and accessibility of symptom management clinical trials for AYA cancer survivors.

Keywords: oncology, adolescent, young adult, qualitative research, surveys and questionnaires

Introduction

The number of adolescent and young adult (AYA, 15–39 years old) cancer survivors living with late effects of cancer treatment is increasing as the five-year relative survival rate for AYAs with cancer is approximately 80%.[1] The most frequent challenges experienced by AYA cancer survivors include physical (e.g., symptoms such as fatigue or pain) and psychological (e.g., anxiety or depression) effects from cancer and/or cancer treatment.[2] AYA cancer survivors

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may experience, on average, nine separate cancer treatment-related symptoms at a moderate or greater severity, and increased symptom severity is associated with further reductions in quality of life.[3] Despite the known negative effects of cancer treatment-related symptoms on quality of life among AYA cancer survivors, few studies have been conducted that specifically test oncology symptom management interventions among AYA cancer survivors.[4–7]

In addition to the lack of oncology symptom management interventions for AYA cancer survivors, several reviews have highlighted the unmet needs and/or challenges faced by AYA cancer survivors pertaining to cancer treatment-related symptom management.[8–10] For example, AYAs report challenges (e.g., need assistance with scheduling, transportation, insurance coordination) related to accessing psychosocial services or counseling,[2,10] peer support groups,[11] or a lack of information/education.[12] However, to our knowledge, few studies[13] have explored AYA cancer survivors' experiences living with and managing cancer treatment-related symptoms using qualitative methodology. A greater understanding of AYA cancer survivors' experiences and barriers to cancer treatment-related symptom assessment and management is needed to begin to develop and/or identify strategies to support AYA cancer survivors with cancer treatment-related symptoms. The purpose of this qualitative study was to explore AYA cancer survivors' experience (e.g., severity, interference, symptom reporting, education, clinical trials, and management) with cancer treatment-related symptoms.

Methods

Design, sample, and setting

AYA cancer survivors were invited to participate in 45-minute semi-structured interviews. Up to 25 interviews were initially planned, but thematic saturation pertaining to participants' perspectives on cancer treatment-related symptoms was obtained after 19 interviews. AYA cancer survivors (15–39 years old) who were at least one month after cancer treatment (e.g., chemotherapy) and self-reported ≥4/10 severity for at least one symptom on the MD Anderson Symptom Inventory were eligible to participate. All participants were recruited from Michigan Medicine or using umhealthresearch.org. All participants were approached via telephone or email. Written informed consent was obtained from all participants, and the study was approved by the University of Michigan IRBMED (HUM00218510). The qualitative methods and results are presented with regard to the Consolidated Criteria for Reporting Qualitative Research guidelines.[14]

Data collection

Participants were emailed a link to complete a demographics questionnaire; questions pertaining to cancer treatment-related symptom importance, assessment, and education; and the MD Anderson Symptom Inventory. The MD Anderson Symptom Inventory measures the severity (13 items) and interference (6 items) of cancer treatment-related symptoms over the past 24 hours (0–10, higher scores represent worse severity or interference).[15] Participants were scheduled to complete the interview via web conferencing or telephone. All interviews were audio recorded by a PhD-prepared nurse scientist (R.K., male) or graduate public health students (L.S. or K.G., female). R.K. had prior experience conducting semi-structured interviews in research settings,[16–18] whereas L.S. and K.G. were new to semi-structured interviewing and received training from R.K. before conducting any interviews. Some participants interacted with the interviewers during the informed consent process before the interview and were aware of the purpose of the interviews. The interviewer and participant were the only individuals present during the interviews. No repeat interviews were conducted and field notes were written by the interviewers as needed. The interview guide is described in Table 1. The questions prompted participants to share their experience with cancer treatment-related symptoms and perceptions of clinical trial participation for cancer treatment-related symptoms. Participants received a $25 gift card for completing the survey and interview.

Qualitative analysis

The audio-recorded interviews were transcribed verbatim by a professional transcription company and verified for accuracy by a study team member (K.G.). The transcripts were not returned to the participants for comments or corrections. The finalized transcripts were imported into NVivo 12 (QSR International Pty Ltd). Inductive content analysis was used to analyze the interview transcripts.[19,20] The principal investigator ([PI] R.K.) and a study team member (K.G.) reviewed the transcripts and interview guide to create a list of initial codes. Next, three transcripts were independently coded using the initial codebook. After three interviews were coded, two study team members (R.K. and K.G.) met to resolve any coding discrepancies and to revise the codebook further. The same process was repeated after three more interviews were coded. After the codebook was finalized, the PI (R.K.) coded the remaining interviews. Subsequently, members of the study team (R.K., L.S., K.G., and E.D.) met as a group to review the coded transcripts to make sense of the data and generate potential major themes and subthemes. The participants did not provide feedback on the themes or subthemes nor received the transcripts for additional comment.

Results

Sample characteristics

Of the 115 AYAs approached about study participation, 24 consented to participate (i.e., no response from 67 individuals and 24 declined participation owing to a lack of interest or time). Ultimately, data from only 19 interviews were included in the analysis as one participant was found to be ineligible (i.e., less than one month after cancer treatment) and four were lost to follow-up. Table 2 describes the demographic characteristics of the participants who completed the interviews. Briefly, on average, participants were approximately 32 years old and 619 days after cancer treatment. Participants were also mainly White (88%) and female (78%), were diagnosed with breast cancer (63%), and learned about cancer treatment-related symptoms via a discussion with their clinician (89%). Participants reported the highest mean scores for the symptoms of fatigue (4.72), distress (3.61), and work-related interference (3.72). The interviews occurred from November 15, 2022, to May 26, 2023, and

Table 1. Semi-Structured Interview Guide Administered to AYA Cancer Survivors

Questions
Tell me what it has been like for you having to live with side effects and symptoms from cancer after treatment?
Probe: What types of symptoms and side effects have you experienced?
Probe: How has it affected your everyday life?
Probe: Can you give me an example of [. . .]?
Probe: Is symptom severity, interference with activities, or frequency most distressing/bothersome?
Probe: Has there been a difference in the severity or type of symptoms you have experienced from the time you were on treatment to the time after completing treatment?
What has been your experience communicating your cancer treatment-related symptoms and side effects to your health care team (e.g., doctors or nurses)?
Probe: How has your health care team been helpful in addressing your symptoms?
Probe: Have you experienced any barriers to addressing your symptoms from cancer treatment with your health care team?
Probe: Would there be a way to make things easier to help cancer survivors communicate cancer treatment-related symptoms to their health care team?
Probe: How did you learn about side effects of cancer treatment? What did your doctor or nurse tell you about cancer treatment-related symptoms?
What kinds of treatments have you tried for the symptoms and side effects from cancer treatment?
Probe: Were the treatments initiated on your own (e.g., self-management) or recommended by your health care team?
Probe: Have the treatments you tried been effective for [symptom]?
Probe: Have you experienced any barriers to using [name of treatment]?
Probe: Looking to the future, what types of treatments would you be interested in trying for the management of cancer treatment-related symptoms?
What has been your experience with clinical trials? Would you consider joining a clinical trial to help manage cancer treatment-related symptoms?
Probe: What are the barriers to adolescent and young adult cancer survivors joining a clinical trial for cancer treatment-related symptom management?
Probe: What would help make joining a clinical trial for cancer treatment-related symptom management appealing for adolescent and young adult cancer survivors?
(After several interviews): Some people have mentioned [. . .]. What do you think about that?
Are there any final comments or anything that we missed in our discussions regarding cancer treatment-related symptoms among AYA cancer survivors?

AYA, adolescent and young adult.

were on average of 16 minutes in length (*range* = 7–23). Four main themes related to AYA cancer survivors' experience with cancer treatment-related symptoms emerged following a review of the transcripts (Table 3).

Cancer treatment-related symptoms negatively affect AYA cancer survivors' quality of life

AYA cancer survivors self-reported cancer treatment-related symptoms, such as lymphedema, fatigue, pain, nausea, sleep disturbance, cognitive impairment, skin rashes, anxiety, peripheral neuropathy, dry eye, depression, and dizziness. The majority of participants reported that cancer treatment-related symptoms negatively impacted the ability to complete day-to-day activities (e.g., school, work, exercise, childcare). A subset of AYA cancer survivors reported that the experience of persistent cancer treatment-related symptoms led to a perception of "feeling older." For example, some participants reported frustration living with symptoms thought to be generally associated with older age (e.g., tiredness, poor memory, mobility issues). Others questioned whether they would be able to complete day-to-day tasks more efficiently without persistent cancer treatment-related symptoms. Furthermore, nearly half of participants revealed that cancer treatment-related symptoms serve as a negative reminder that their cancer may recur in the future. For example, one participant stated, "I think it's hard not to worry every pain or symptom that you feel like, 'I have a headache, is the cancer coming back?' So it definitely causes stress and I think sadness too." In particular, several participants reported that the anxiety and/or depression associated with fear of cancer recurrence were worse than the day-to-day physical side effects of cancer treatment.

AYA cancer survivors' attitudes and feelings about communicating cancer treatment-related symptom concerns to clinicians

The majority of AYA cancer survivors reported positive experiences communicating cancer treatment-related symptom concerns to their health care team. Perceived facilitators to successful patient–clinician communication about symptom concerns included (1) AYAs viewed the clinician as approachable, friendly, and/or warm; (2) AYAs felt that their concerns were being taken seriously by the clinician (e.g., receptive, attentive); (3) AYAs reported that the clinician spent an adequate amount of time to educate about potential symptoms and discuss concerns; and (4) AYAs thought the clinicians were easy to contact (e.g., portal messaging). However, participants also reported several barriers that

Table 2. Demographic Characteristics of the Enrolled Participants ($N = 19$)

Patient-reported characteristics (n = *18*)[a]	*Frequency or mean*
Age at baseline	
Mean (*SD, Range*)	31.83 (6.52, 18–39)
Gender	
Female	14 (78%)
Male	4 (22%)
Race	
Black or African American	1 (6%)
White	17 (94%)
Ethnicity	
Hispanic or Latino	1 (6%)
Not Hispanic or Latino	17 (94%)
Education	
Some college or technical training	4 (22%)
University undergraduate degree	6 (33%)
University postgraduate degree	8 (44%)
Marital status	
Single	7 (39%)
Married/Partnered	11 (61%)
Employment status	
Working full-time	9 (50%)
Working part-time	3 (17%)
Student	2 (11%)
Not working	4 (22%)
Prior research study participation	
Zero	4 (22%)
One	5 (28%)
Two	5 (28%)
Three or more	4 (22%)
MD Anderson Symptom Inventory severity	*Mean* (*SD, Range*)
Pain	3.17 (3.24, 0–10)
Fatigue	4.72 (2.89, 1–9)
Nausea	1.78 (3.14, 0–10)
Disturbed sleep	3.17 (2.9, 0–8)
Distress	3.61 (3.07, 0–9)
Shortness of breath	1.72 (2.76, 0–8)
Problem remembering	3.0 (2.11, 0–7)
Lack of appetite	1.78 (2.58, 0–7)
Drowsiness	3.5 (3.17, 0–9)
Dry mouth	1.83 (2.96, 0–8)
Sadness	3.0 (3.05, 0–8)
Vomiting	1.11 (2.22, 0–7)
Numbness or tingling	2.56 (2.28, 0–7)
MD Anderson Symptom Inventory interference	*Mean* (*SD, Range*)
General activity	3 (2.91, 0–10)
Mood	3.39 (2.91, 0–10)
Work (including work around the house)	3.72 (2.87, 0–10)
Relations with other people	2.5 (2.75, 0–9)
Walking	1.44 (2.25, 0–7)
Enjoyment of life	3.11 (2.54, 0–9)

(*continued*)

Table 2. (Continued)

Patient-reported characteristics (n = *18*)[a]	*Frequency or mean*
Where did you learn information about cancer treatment-related symptoms or side effects before beginning cancer treatment?[b]	
Doctor or nurse	16 (89%)
Friend or family member who had cancer	5 (28%)
Internet search	12 (67%)
I did not learn about cancer treatment-related symptoms before beginning cancer treatment	1 (6%)
How did your doctor or nurse present information about cancer treatment-related symptoms?[b]	
Discussion with your doctor or nurse	16 (89%)
Paper education sheets	13 (72%)
What would be the best way for you to learn about cancer treatment-related symptoms?[b]	
Discussion with your doctor or nurse	16 (89%)
Paper education sheets	14 (78%)
Videos	6 (33%)
Internet resources	12 (67%)
When would be the best time to receive information about cancer treatment-related symptoms?	
Before you begin cancer treatment	16 (89%)
When you first experience cancer treatment-related symptoms	2 (11%)
What would be the best way for you to report cancer treatment-related symptoms to your doctor or nurse?[b]	
Complete surveys about cancer treatment-related symptoms before your appointment (e.g., in-person in waiting room)	4 (22%)
Complete surveys about cancer treatment-related symptoms before your appointment (e.g., via email or text before appointment)	12 (67%)
Tell your doctor or nurse verbally during the appointment	11 (61%)
Medical record abstraction ($n = 19$)	
Cancer type	
Leukemia	4 (21%)
Lymphoma	2 (11%)
Melanoma	1 (5%)
Breast	12 (63%)
Cancer stage	
Stage I	8 (42%)
Stage II	3 (16%)
Stage III	3 (16%)
Metastatic	1 (5%)
Unknown	4 (21%)

(*continued*)

Table 2. (Continued)

Patient-reported characteristics (n = *18*)[a]	*Frequency or mean*
Prior treatment type[b]	
Chemotherapy	15 (79%)
Immunotherapy	2 (11%)
Surgery	12 (63%)
Radiation	9 (47%)
Hormone therapy	8 (42%)
Autologous stem cell transplant	2 (11%)
Days from cancer treatment to the time of consent	
Mean (*SD*)	618.53 (401.29)

[a]One participant did not complete the patient-reported outcome measures.
[b]Sum of percentages equals greater than 100% as the question or category allowed for multiple selections.
SD, standard deviation.

hindered effective communication with their clinicians about cancer treatment-related symptom concerns. Primarily, several participants reported that clinicians may downplay reports of cancer treatment-related symptoms with responses such as, "Oh, yeah, that's going to happen," "There's nothing we can do," "Well, that's not usually what happens," or "Oh, it'll let up." Subsequently, the AYA cancer survivors felt as if their concerns were not being taken seriously by the clinician. In response to clinicians downplaying symptom concerns, a few participants articulated the tension of taking the clinician at their word and trusting versus further advocating about their symptom concerns to the clinicians (e.g., need for referral or further treatment). Secondarily, some AYA cancer survivors felt as if they did not have the right to complain to the clinician about symptom concerns now that their cancer has been treated (e.g., do not want to be perceived as ungrateful or burden clinician with symptom concerns). A few participants also reported a lack of preparedness in how to identify or manage cancer treatment-related symptoms after active cancer treatment was completed (e.g., learned as the symptoms arose or symptom and side effect education was more focused on the active cancer treatment phase). Moving forward, AYAs self-reported several strategies to improve patient–clinician communication: (1) taking the time to discuss symptoms in person (e.g., clinicians do not always read patient-reported outcomes surveys before visit), (2) enlisting a liaison/advocate to facilitate patient–clinician communication and follow-up treatment, and (3) education about what to report/communicate to clinicians about cancer treatment-related symptoms.

AYA cancer survivors are interested in oncology symptom management clinical trials, but logistical challenges are barriers to participation

All participants were interested in participating in oncology symptom management clinical trials in the future, with some AYAs having prior clinical research experience. A primary motivating factor to clinical trial participation was whether the clinical trial was personally relevant. Specifically, a clinical trial was personally relevant if the proposed clinical trial goal was viewed as helping other AYA cancer survivors in the future or potentially efficacious for a symptom that the participant was actively experiencing. A primary barrier to AYA cancer survivors participating in oncology symptom management trials was lack of awareness of available clinical trials (e.g., AYA cancer survivors do not know how to seek out such trials and oncologists may not be aware). The participants also identified several logistical factors that may hinder oncology symptom management clinical trial participation for AYA cancer survivors more generally: (1) flexible scheduling to accommodate work, school, or distance from the cancer center (e.g., availability on weekend, combined research and clinical appointments, virtual surveys, or intervention delivery), (2) childcare support, (3) lack of financial incentives, and (4) research staff is not matched to sex or age of participant. Lastly, one participant stated that a barrier to clinical trial participation for adolescents might be that some people want to get away from all oncology-related activities after cancer treatment.

AYA cancer survivors are interested in nonpharmacological treatments for symptom management

AYA cancer survivors self-reported using a variety of nonpharmacological treatments for oncology symptom management that were recommended by the health care team or self-initiated. The AYA cancer survivors' health care teams facilitated referrals to physical therapy or psychological therapy for oncology symptom management. Nearly half of AYA cancer survivors self-reported the initiation of several nonpharmacological treatments for oncology symptom management, including diet changes, exercise, heat therapy (e.g., sauna), yoga, acupuncture, cannabis, vitamins and supplements, and massage. Self-reported barriers to nonpharmacological treatment use among AYA cancer survivors for oncology symptom management included time and cost (e.g., not covered by insurance). The majority of AYA cancer survivors also identified an interest in nonpharmacological treatments for oncology symptom management in the future (e.g., physical therapy, group therapy, anxiety support program, yoga, acupuncture, cupping, vitamins/supplements, massages, mindfulness, and relaxation).

Discussion

The results of this qualitative study highlighted the impact of cancer treatment-related symptoms on AYA cancer survivors' day-to-day life. Consistent with prior reports, AYA cancer survivors experienced a multitude of side effects following cancer treatment that negatively affected physical functioning and/or quality of life.[3,13] Furthermore, although numerous studies demonstrate that increased anxiety, depression, and stress are associated with increased fear of cancer recurrence,[21–23] our data also suggest that the physical symptoms experienced by participants (e.g., fatigue, sleep disturbance, cognitive impairment) cultivated ongoing psychological concerns, such as "feeling older" or increased fear of cancer recurrence. Severe fear of cancer recurrence (up to $\sim$62% of AYAs)[23] is significantly associated with increased use of health care (e.g., consultations and medications)[24,25] and excessive symptom monitoring.[26] However, there are currently no recommended interventions to support psychological distress in AYAs with cancer.[7] Future interventions may target both physical (e.g., fatigue)[27] and

Table 3. Summary of Themes and Associated Exemplar Quotes from the Semi-Structured Interviews with AYA Cancer Survivors

Themes	*Exemplar quotes*
Cancer treatment-related symptoms negatively affect AYA cancer survivors' quality of life	I think people will talk about really common side effects, but one of the things that I was not expecting at all was, you almost feel like doing all that treatment ages your mind and your body and you feel older than you did before, which is just sometimes you just get exhausted in ways that you never did before. (8) I guess they're a constant reminder that you had cancer and it could come back at any time. So, that's difficult mentally to have that, that's harder for me than the physical aspect. (13)
AYA cancer survivors' attitudes and feelings about communicating cancer treatment-related symptom concerns to clinicians	I feel like there wasn't really much they could do for me for my nausea and fatigue besides just saying, "Oh, it'll let up." (12) I don't know if it's a female thing or a cancer patient thing or a younger person thing, but I felt personally like I didn't have the right to complain. Like, "It doesn't matter how bad it gets or that my feet are tingly. We're not doing it for that." And, "My healthcare team doesn't want to hear about it." That's how I felt. (9) There's a difference between you know something's wrong and advocating for yourself, versus taking people at their word and trusting. And I think that that's a very delicate balance throughout this process. So as far as barriers, I think it's just what level of trust do you have with your healthcare providers? And is it something that you as the patient should be advocating more for, or is it like, "Okay, this is something, I'm going to take them at their word and I trust them?" (14)
AYA cancer survivors are interested in oncology symptom management clinical trials, but logistical challenges are barriers to participation	I think just showing how it would benefit them in their current state, how it would benefit some symptoms they might experience. (11) Just as far as scheduling, having things available in the evenings or weekends, having childcare. I know this is just a dream scenario, but having childcare would be amazing because there's definitely ones that I'm like, "I can't participate in that because someone has to watch my daughter." (1) Not being asked or not knowing about them or not knowing how they could benefit us. Some barriers just might be time. If they're in school or they're working, it might be difficult to dedicate their time to something else. (6)
AYA cancer survivors are interested in nonpharmacological treatments for symptom management	During treatment, you get to a point where you're taking medication to treat the side effects of other medications to treat the side effects of other medications. So you take chemo and then you're in pain because of the chemo. So you take pain medication and then you have side effects on the pain medication that you have to take some sort of medication for. (17)

Numbers represent the participant ID associated with the quote.

psychological predictors[22] of increased fear of cancer recurrence. For example, a mindfulness-based cognitive therapy intervention was found to significantly improve psychological distress ($p < 0.001$), fear of cancer recurrence ($p < 0.05$), and fatigue ($p < 0.01$) among adults with cancer relative to a usual care control ($n = 74$).[28]

Most AYA cancer survivors reported that their cancer treatment-related concerns were discussed and addressed by clinicians, but effective communication was hindered when AYAs perceived that their cancer treatment-related symptom concerns were not taken seriously by the clinician. The facilitators and barriers to AYA–clinician communication about cancer treatment-related symptoms may be discussed within the context of the eight interdependent functions of communication in AYA oncology,[29,30] based on the National Cancer Institute framework of patient-centered oncology communication.[31,32] For example, the principle, "building relationships," was observed when AYAs were more likely to report positive cancer treatment-related symptom discussions with clinicians who were viewed as warm, friendly, or approachable and provided timely follow-up on symptom concerns. Similarly, the principles of "exchanging information" and "providing validation" were promoted when AYAs felt that clinicians took the time to address questions about cancer treatment-related symptoms and took their symptom concerns seriously. As for "enabling family self-management," AYAs who reported the receipt of symptom education from clinicians appeared to have more positive symptom management discussions than those who were unprepared for cancer treatment-related symptoms. On the contrary, the principles of "managing uncertainty," "responding to emotions," "making decisions," and "supporting hope" were not exemplified when AYAs' symptom concerns were downplayed by clinicians, or no treatment options were offered in response to cancer treatment-related symptom concerns.

Despite the potential gaps in symptom management communication between AYA cancer survivors and clinicians, few studies have been conducted to improve patient–clinician symptom assessment and communication in AYA oncology.[33] A potential barrier to the development of interventions to improve AYA–clinician symptom assessment and communication is that there are few measures specifically tailored to the unique needs and symptom experiences of AYA cancer

survivors.[34] Moving forward, as recommended by the National Cancer Institute's Clinical Trials Network AYA PRO Task Force, future intervention studies should consider using the AYA patient-reported outcomes Core Battery to assess for cancer treatment-related symptoms.[35] The recommended AYA core battery includes PROMIS short forms to assess for health-related quality of life (i.e., depression, anxiety, fatigue, physical function, pain interference, emotional support, cognitive function, and pain intensity) and PRO-CTCAE items to assess for study-specific adverse events.[35] The measures were selected to be included in the core battery because they were relevant to the AYA symptom experience, validated, available without cost, translated in multiple languages, and applicable to different cancer types.[35] The implementation of patient-reported outcome measures that capture the symptom experiences of AYA cancer survivors will promote the timely identification and management of cancer treatment-related symptoms.

Clinical trial enrollment among AYA cancer survivors is poor,[36–38] and less is known about facilitators and barriers to enrolling AYA cancer survivors in oncology symptom management clinical trials. AYAs reported increasing the flexibility in the delivery of the planned intervention and associated study visits or measures and providing incentives as strategies to facilitate AYA enrollment into oncology symptom management clinical trials. Consistent with prior data,[39] participants also reported lack of awareness of applicable research trials among AYAs and clinicians as a barrier to clinical trial participation. Moving beyond system level and logistical barriers, a primary motivation for AYA participation in oncology symptom management clinical trials is the level of interest in the research objective for themselves or future AYA cancer survivors. Future research is needed to identify and integrate AYAs' preferences into oncology symptom management clinical trial design to potentially increase AYA recruitment and retention to such trials. One such method to accomplish this future research objective is to conduct a choice-based conjoint analysis where participants are asked to choose the most desirable oncology symptom management clinical trial design factors.[40] For instance, a choice-based conjoint analysis was conducted to identify cancer survivors' preferences for participation in chemotherapy-induced peripheral neuropathy clinical trials.[41] Results revealed that type of treatment (i.e., highest preference for nonpharmacological) and length of study visits were the two most important factors for cancer survivors to consider when deciding whether to enroll in chemotherapy-induced peripheral neuropathy clinical trials.[41]

Implications for practice

AYA cancer survivors identified several barriers to communicating cancer treatment-related symptoms with clinicians. To address communication barriers, clinicians may consider integrating the following patient-centered communication principles when discussing cancer treatment-related symptoms with AYA cancer survivors: (1) interpersonal relationships (e.g., clinicians demonstrate commitment, empathy, and hopefulness), (2) informational preferences (e.g., prepare for what is to come and help to understand current situation), and (3) delivery of treatment, resources, and medical care (e.g., providing referrals and resources).[42] In addition, given AYA cancer survivors' interest in nonpharmacological treatments, clinicians should discuss the availability of integrative oncology treatments at the cancer center or in the community with AYA cancer survivors suffering from cancer treatment-related symptoms, although the evidence base is still emerging for many integrative oncology treatments for various symptoms (e.g., pain,[43] anxiety, or depression).[44]

Limitations

There are several limitations to the findings of this study. The external generalizability is limited as the sample was homogeneous with regard to sex, race, institution, and ethnicity. As a result, participants' experiences and preferences with regard to cancer treatment-related symptoms may have differed if the sample consisted of greater diversity. Furthermore, the results may not be generalizable specifically to adolescents as no adolescents (15–17 years old) were interviewed. The interview questions were also designed to describe AYA cancer survivors' experiences with cancer treatment-related symptoms overall, with no emphasis on specific symptoms. Thus, the study results may not be applicable to all cancer treatment-related symptoms. Finally, we were unable to document participants' nonverbal expressions during the interviews, as the interviews were conducted using videoconferencing or telephone only.

Conclusion

The results of the interviews revealed that AYA cancer survivors experience a myriad of cancer treatment-related symptoms that negatively affect physical and psychological functioning. In particular, the physical symptoms experienced by AYA cancer survivors appeared to drive increased psychological distress. While it is not surprising that cancer treatment-related symptoms negatively affected quality of life in this population, the results also revealed several barriers in patient–clinician communication and clinical trial participation that may be explored to decrease AYA cancer survivors' symptom distress and improve quality of life.

Authors' Contributions

R.K.: Investigation, methodology, formal analysis, and writing—original draft. K.G.: Formal analysis, data curation, and investigation. L.S.: Formal analysis and investigation. D.B.: Supervision and methodology. All authors contributed to writing—reviewing and editing and conceptualization of the study.

Author Disclosure Statement

No competing financial interests exist.

Conflict of Interest Statement

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